

1 Summary

Model (Run-name): model3 (test)

Iteration: 2

TAC: 58798.98902361891

Took: 8.319321155548096s

Total Time: 11.229872465133667s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

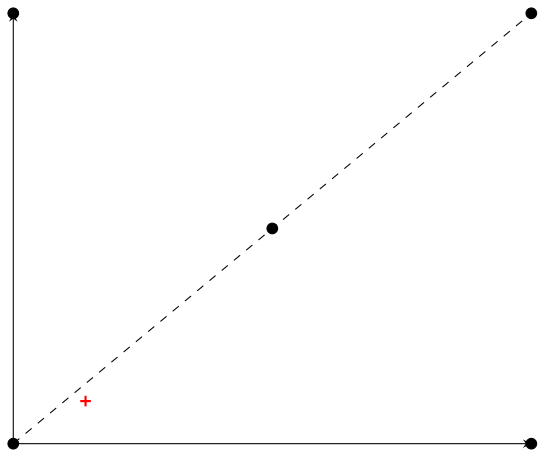
2 Results

Variable	Value	Variable	Value	Variable	Value
$z_{2,2,2}$	1.0	$dt_{3,5,2}$	22.3	$q_{3,5,2}$	1013.83999999999855
$z_{2,5,2}$	1.0	$dt_{3,5,3}$	13.545945945945547	$q_{4,3,2}$	1539.72000000000003
$z_{3,1,2}$	1.0	$dt_{4,3,2}$	50.0	$q_{5,4,2}$	1634.85
$z_{3,5,2}$	1.0	$dt_{4,3,3}$	111.1	$q_{cu,1}$	586.95999999999999
$z_{4,3,2}$	1.0	$dt_{5,4,2}$	22.2	$q_{cu,3}$	610.43999999999838
$z_{5,4,2}$	1.0	$dt_{5,4,3}$	24.33559322033827	$q_{cu,5}$	724.55999999999783
$z_{cu,1}$	1.0	$dt_{cu,1}$	120.0	$t_{1,1}^{(H)}$	160.0
$z_{cu,3}$	1.0	$dt_{cu,3}$	66.84594594594485	$t_{1,2}^{(H)}$	160.0
$z_{cu,5}$	1.0	$dt_{cu,5}$	66.53559322033776	$t_{1,3}^{(H)}$	160.0
$A_{2,2,2}^{\beta}$	3.338874959957588	$f_{2,2,2}^H$	5.825472547254725	$t_{2,1}^{(H)}$	248.9
$A_{2,5,2}^{\beta}$	0.557924632432702	$f_{2,5,2}^H$	4.774527452745428	$t_{2,2}^{(H)}$	248.9
$A_{3,1,2}^{\beta}$	2.2768832383260187	$f_{3,1,2}^H$	4.7175667287399135	$t_{2,3}^{(H)}$	137.8
$A_{3,5,2}^{\beta}$	1.1160166994093859	$f_{3,5,2}^H$	6.2932340161389355	$t_{3,1}^{(H)}$	226.7
$A_{4,3,2}^{\beta}$	2.4893962060607	$f_{4,3,2}^H$	12.6	$t_{3,2}^{(H)}$	226.7
$A_{5,4,2}^{\beta}$	6.27528091057471	$f_{5,4,2}^H$	12.264441110277584	$t_{3,3}^{(H)}$	106.84594594594485
$A_{cu,1}^{\beta}$	2.156787801780001	$f_{2,2,2}^C$	6.1	$t_{4,1}^{(H)}$	271.1
$A_{cu,3}^{\beta}$	2.9483518834332667	$f_{2,5,2}^C$	4.774527452745428	$t_{4,2}^{(H)}$	271.1
$A_{cu,5}^{\beta}$	3.3568468435001897	$f_{3,1,2}^C$	7.6	$t_{4,3}^{(H)}$	148.9
$A_{2,2,2}$	18.89081402008292	$f_{3,5,2}^C$	9.125472547254594	$t_{5,1}^{(H)}$	198.9
$A_{2,5,2}$	4.0106608195978595	$f_{4,3,2}^C$	8.4	$t_{5,2}^{(H)}$	198.9
$A_{3,1,2}$	13.737212823561322	$f_{5,4,2}^C$	17.3	$t_{5,3}^{(H)}$	106.53559322033776
$A_{3,5,2}$	8.941176470587976	$RecLMTD_{2,2,2}$	0.024809863749124014	$t_{1,1}^{(C)}$	160.0
$A_{4,3,2}$	19.92063386984238	$RecLMTD_{2,5,2}$	0.017663772783495423	$t_{1,2}^{(C)}$	160.0
$A_{5,4,2}$	51.43476571725509	$RecLMTD_{3,1,2}$	0.0153639880263515	$t_{1,3}^{(C)}$	60.0
$A_{cu,1}$	7.5275930538960525	$RecLMTD_{3,5,2}$	0.030877760700829676	$t_{2,1}^{(C)}$	221.7
$A_{cu,3}$	7.499856372317694	$RecLMTD_{4,3,2}$	0.010997154540673845	$t_{2,2}^{(C)}$	221.7
$A_{cu,5}$	9.832103793393394	$RecLMTD_{5,4,2}$	0.026742239875014118	$t_{2,3}^{(C)}$	115.6
$dt_{2,2,2}$	27.2	$RecLMTD_{cu,1}$	0.010901005342462312	$t_{3,1}^{(C)}$	221.1
$dt_{2,2,3}$	22.2	$RecLMTD_{cu,3}$	0.020102667571323306	$t_{3,2}^{(C)}$	221.1
$dt_{2,5,2}$	44.5	$RecLMTD_{cu,5}$	0.02285016861537679	$t_{3,3}^{(C)}$	37.8
$dt_{2,5,3}$	44.5	$q_{2,2,2}$	647.2099999999999	$t_{4,1}^{(C)}$	176.7
$dt_{3,1,2}$	66.7	$q_{2,5,2}$	530.45	$t_{4,2}^{(C)}$	176.7
$dt_{3,1,3}$	46.845945945945495	$q_{3,1,2}$	760.0	$t_{4,3}^{(C)}$	82.2

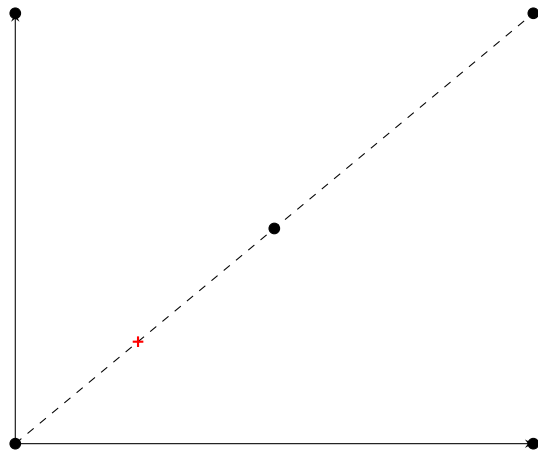
Variable	Value	Variable	Value
$t_{5,1}^{(C)}$	204.4	$b_{2,2,2}^{C,\text{out}}$	1352.37
$t_{5,2}^{(C)}$	204.4	$b_{2,5,2}^{C,\text{out}}$	975.9134113411656
$t_{5,3}^{(C)}$	93.3	$b_{3,1,2}^{C,\text{out}}$	1216.0
$t_{2,2,2}^H$	187.8424528301887	$b_{3,5,2}^{C,\text{out}}$	1865.246588658839
$t_{2,5,2}^H$	198.85754716980975	$b_{4,3,2}^{C,\text{out}}$	1857.2400000000002
$t_{3,1,2}^H$	65.6	$b_{5,4,2}^{C,\text{out}}$	3056.91
$t_{3,5,2}^H$	65.6		
$t_{4,3,2}^H$	148.9		
$t_{5,4,2}^H$	65.6		
$t_{2,2,2}^C$	221.7		
$t_{2,5,2}^C$	131.46187050359833		
$t_{3,1,2}^C$	160.0		
$t_{3,5,2}^C$	166.23812949640183		
$t_{4,3,2}^C$	221.1		
$t_{5,4,2}^C$	176.7		
$b_{2,2,2}^{H,\text{in}}$	1449.960117011701		
$b_{2,5,2}^{H,\text{in}}$	1188.3798829883372		
$b_{3,1,2}^{H,\text{in}}$	1069.4723774053382		
$b_{3,5,2}^{H,\text{in}}$	1426.6761514586974		
$b_{4,3,2}^{H,\text{in}}$	3415.8599999999997		
$b_{5,4,2}^{H,\text{in}}$	2439.3973368342176		
$b_{2,2,2}^{H,\text{out}}$	802.750117011701		
$b_{2,5,2}^{H,\text{out}}$	657.9298829883201		
$b_{3,1,2}^{H,\text{out}}$	309.4723774053383		
$b_{3,5,2}^{H,\text{out}}$	412.8361514587142		
$b_{4,3,2}^{H,\text{out}}$	1876.1399999999996		
$b_{5,4,2}^{H,\text{out}}$	804.5473368341537		
$b_{2,2,2}^{C,\text{in}}$	705.16		
$b_{2,5,2}^{C,\text{in}}$	445.4634113411484		
$b_{3,1,2}^{C,\text{in}}$	456.0		
$b_{3,5,2}^{C,\text{in}}$	851.4065886588536		
$b_{4,3,2}^{C,\text{in}}$	317.52		
$b_{5,4,2}^{C,\text{in}}$	1422.0600000000002		

3 *RecLMTD* Tangent Points

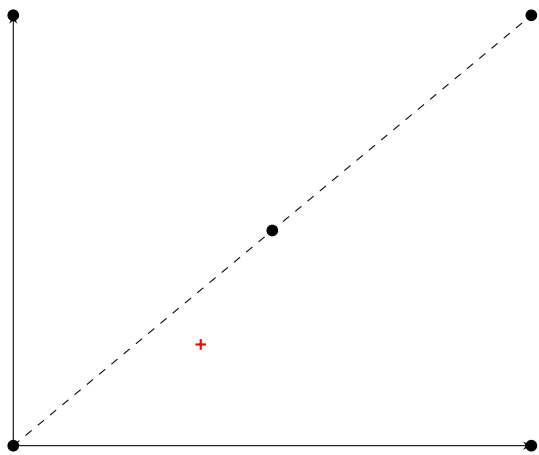
- (2, 2, 2)
 - Value: 0.024810
 - Error: 0.015815073135620
 - Relative error: 0.389294712764435
- (2, 5, 2)
 - Value: 0.017664
 - Error: 0.004808137328864
 - Relative error: 0.213962111134454
- (3, 1, 2)
 - Value: 0.015364
 - Error: 0.002432905225603
 - Relative error: 0.136703928666652
- (3, 5, 2)
 - Value: 0.030878
 - Error: 0.026067211881049
 - Relative error: 0.457761426499381
- (4, 3, 2)
 - Value: 0.010997
 - Error: 0.002070074448155
 - Relative error: 0.158417247445855
- (5, 4, 2)
 - Value: 0.026742
 - Error: 0.016265824808696
 - Relative error: 0.378204063082537
- $cu, 1$
 - Value: 0.010901
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 3$
 - Value: 0.020103
 - Error: 0.001104587178696
 - Relative error: -0.058142041504581
- $cu, 5$
 - Value: 0.022850
 - Error: 0.003804181501617
 - Relative error: -0.199736641576779



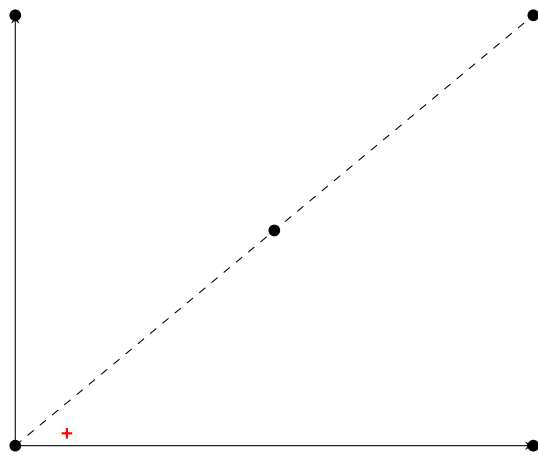
(a) *RecLMTD* - (2, 2, 2)



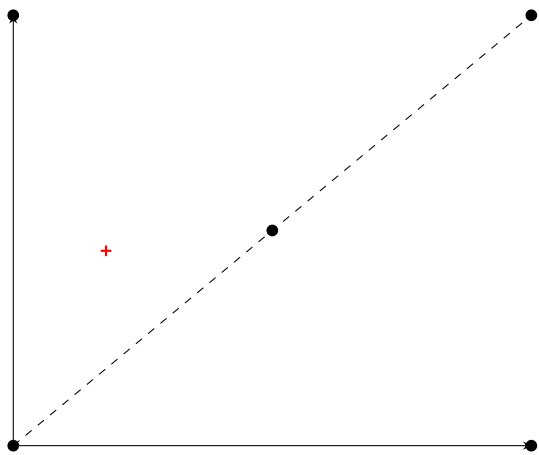
(b) *RecLMTD* - (2, 5, 2)



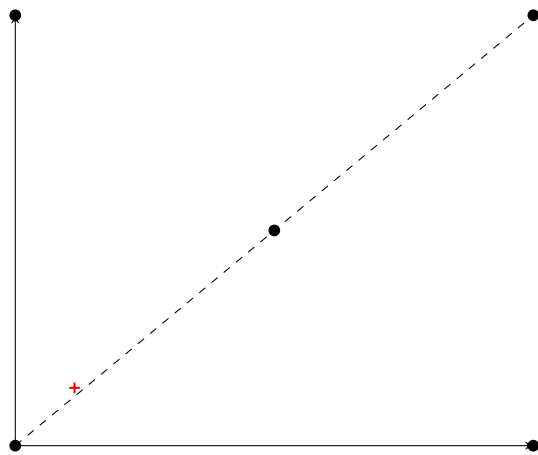
(c) *RecLMTD* - (3, 1, 2)



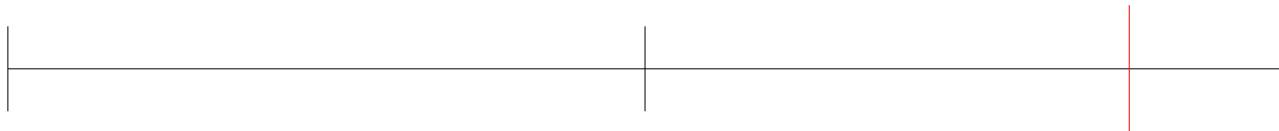
(d) *RecLMTD* - (3, 5, 2)



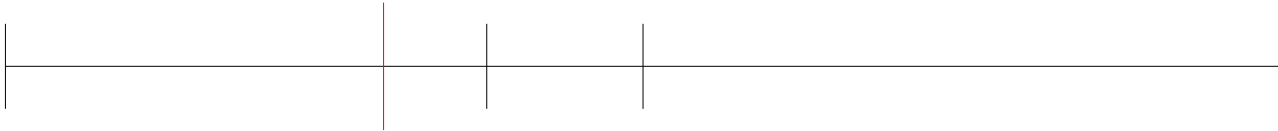
(e) *RecLMTD* - (4, 3, 2)



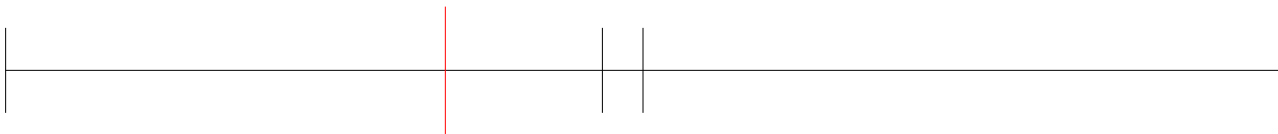
(f) *RecLMTD* - (5, 4, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - $cu, 3$



(c) *RecLMTD* - $cu, 5$

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (2, 2, 2)
 - Value: 1449.960117011701
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 5, 2)
 - Value: 1188.3798829883372
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 1.9133080145337968e-16
- (3, 1, 2)
 - Value: 1069.4723774053382
 - Absolute error: 2.2737367544323206e-13
 - Relative error: -2.1260359804230424e-16
- (3, 5, 2)
 - Value: 1426.6761514586974
 - Absolute error: 6.821210263296962e-13
 - Relative error: 4.781190360771542e-16
- (4, 3, 2)
 - Value: 3415.8599999999997
 - Absolute error: 4.547473508864641e-13
 - Relative error: -1.3312821687260722e-16
- (5, 4, 2)
 - Value: 2439.3973368342176
 - Absolute error: 5.9117155615240335e-12
 - Relative error: 2.423432817712308e-15

4.2 $b^{H,\text{out}}$

- (2, 2, 2)
 - Value: 802.750117011701
 - Absolute error: 291.5209351595539
 - Relative error: -0.26640651288464356
- (2, 5, 2)
 - Value: 657.9298829883201
 - Absolute error: 291.5209351595554
 - Relative error: -0.3070416387951877
- (3, 1, 2)

- Value: 309.4723774053383
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 2)
 - Value: 412.8361514587142
 - Absolute error: 5.684341886080802e-14
 - Relative error: 1.3769002220362153e-16
- (4, 3, 2)
 - Value: 1876.1399999999996
 - Absolute error: 4.547473508864641e-13
 - Relative error: -2.4238455066597593e-16
- (5, 4, 2)
 - Value: 804.5473368341537
 - Absolute error: 5.5706550483591855e-11
 - Relative error: -6.923961827130022e-14

4.3 $b^{C,in}$

- (2, 2, 2)
 - Value: 705.16
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 5, 2)
 - Value: 445.4634113411484
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 1, 2)
 - Value: 456.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 2)
 - Value: 851.4065886588536
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 3, 2)
 - Value: 317.52
 - Absolute error: 0.0
 - Relative error: 0.0

- (5, 4, 2)
 - Value: 1422.0600000000002
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (2, 2, 2)
 - Value: 1352.37
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 5, 2)
 - Value: 975.9134113411656
 - Absolute error: 348.2451016324709
 - Relative error: 0.554823457303578
- (3, 1, 2)
 - Value: 1216.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 5, 2)
 - Value: 1865.246588658839
 - Absolute error: 348.2451016324701
 - Relative error: 0.2295614767755444
- (4, 3, 2)
 - Value: 1857.2400000000002
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 1.2242557528549464e-16
- (5, 4, 2)
 - Value: 3056.91
 - Absolute error: 0.0
 - Relative error: 0.0



$$(a) \ t^{(H)} - (2, 2)$$



$$(a) \ t^{(H)} - (2, 2)$$



$$(a) \ t^{(H)} - (3, 2)$$



$$(a) \ t^{(H)} - (3, 2)$$



$$(a) \ t^{(H)} - (4, 2)$$



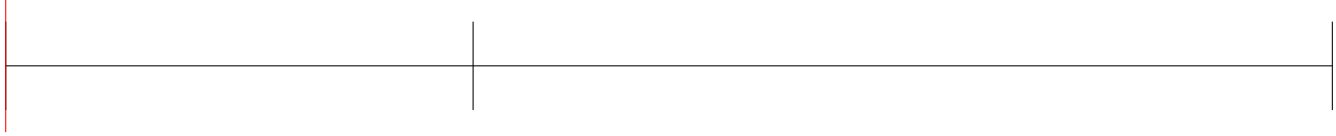
$$(a) \ t^{(H)} - (5, 2)$$



$$(a) \ t^{(H)} - (2, 2, 2)$$



$$(a) \ t^{(H)} - (2, 5, 2)$$



$$(a) \ t^{(H)} - (3, 1, 2)$$



$$(a) \ t^{(H)} - (3, 5, 2)$$



$$(a) \ t^{(H)} - (4, 3, 2)$$



$$(a) \ t^{(H)} - (5, 4, 2)$$



$$(a) \ t^{(C)} - (2, 3)$$



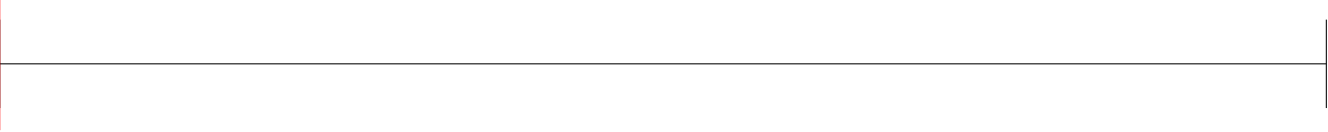
$$(a) \ t^{(C)} - (5, 3)$$



$$(a) \ t^{(C)} - (1, 3)$$



$$(a) \ t^{(C)} - (5, 3)$$



$$(a) \ t^{(C)} - (3, 3)$$



$$(a) \ t^{(C)} - (4, 3)$$



$$(a) \ t^{(C)} - (2, 2, 2)$$



$$(a) \ t^{(C)} - (2, 5, 2)$$



$$(a) \ t^{(C)} - (3, 1, 2)$$



$$(a) \ t^{(C)} - (3, 5, 2)$$



$$(a) \ t^{(C)} - (4, 3, 2)$$



$$(a) \ t^{(C)} - (5, 4, 2)$$

5 Area Breakpoints

5.1 A

- $(2, 2, 2)$
 - Value: 18.890814
 - Error: 0.0000000000000082
 - Relative error: 0.0000000000000004
- $(2, 5, 2)$
 - Value: 4.010661
 - Error: 7.012572442761138
 - Relative error: 0.636162936577506
- $(3, 1, 2)$
 - Value: 13.737213
 - Error: 0.0000000000000020
 - Relative error: 0.0000000000000001
- $(3, 5, 2)$
 - Value: 8.941176
 - Error: 27.888363422269329
 - Relative error: 0.757228124581539
- $(4, 3, 2)$
 - Value: 19.920634
 - Error: 0.0000000000000366
 - Relative error: 0.0000000000000018
- $(5, 4, 2)$
 - Value: 51.434766
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000

5.2 A_{CU}

- $cu, 1$
 - Value: 7.527593
 - Error: 0.0000000000000040
 - Relative error: 0.0000000000000005
- $cu, 3$
 - Value: 7.499856
 - Error: 6.937169971492040
 - Relative error: 0.480512385742549
- $cu, 5$

- Value: 9.832104
- Error: 9.645917585379443
- Relative error: 0.495220607771361



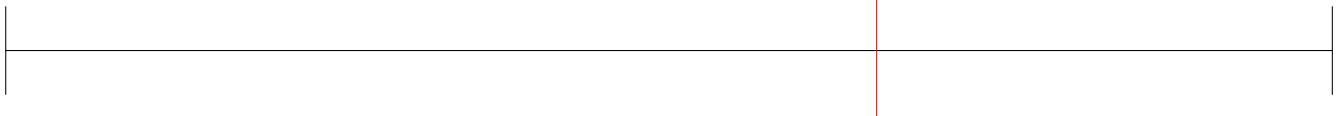
(a) $A^\beta - (2, 2, 2)$



(a) $A^\beta - (2, 5, 2)$



(a) $A^\beta - (3, 1, 2)$



(a) $A^\beta - (3, 5, 2)$



(a) $A^\beta - (4, 3, 2)$



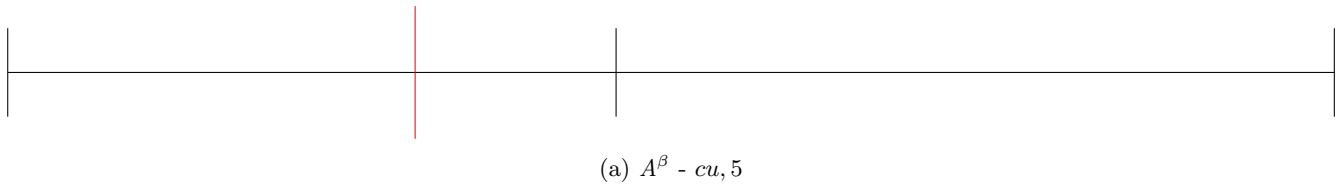
(a) $A^\beta - (5, 4, 2)$



(a) $A^\beta - cu, 1$



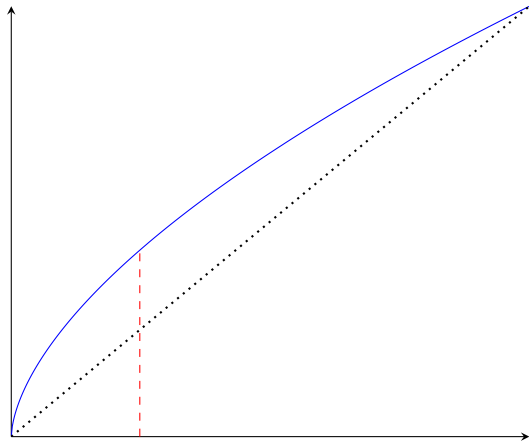
(a) $A^\beta - cu, 3$



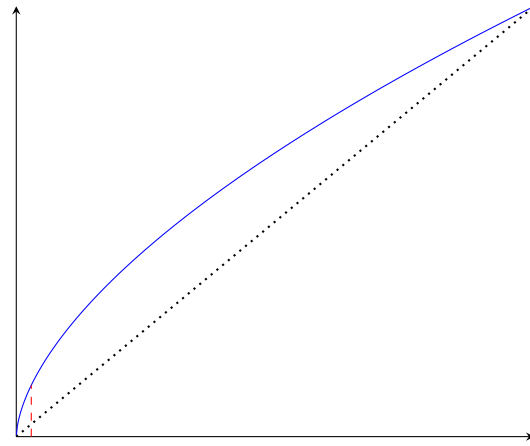
6 A^β Breakpoints

- (2, 2, 2)
 - q : 18.890814
 - $\hat{q}^\beta$: 3.338875
 - q^β : 5.831099
 - Breakpoints: [0, 76.14235294117645]
 - Error: 2.492223886234486
 - Relative error: 0.427402098982082
- (2, 5, 2)
 - q : 4.010661
 - $\hat{q}^\beta$: 0.557925
 - q^β : 2.301069
 - Breakpoints: [0, 138.54823529411763]
 - Error: 1.743143941477134
 - Relative error: 0.757536720652914
- (3, 1, 2)
 - q : 13.737213
 - $\hat{q}^\beta$: 2.276883
 - q^β : 4.816584
 - Breakpoints: [0, 89.41176470588236]
 - Error: 2.539701175203998
 - Relative error: 0.527282604675184
- (3, 5, 2)
 - q : 8.941176
 - $\hat{q}^\beta$: 1.116017
 - q^β : 3.722518
 - Breakpoints: [0, 181.68117647058827]
 - Error: 2.606501244937639
 - Relative error: 0.700198436624286
- (4, 3, 2)
 - q : 19.920634
 - $\hat{q}^\beta$: 2.489396
 - q^β : 6.019798
 - Breakpoints: [0, 181.14352941176475]
 - Error: 3.530401429830401
 - Relative error: 0.586465134439324
- (5, 4, 2)

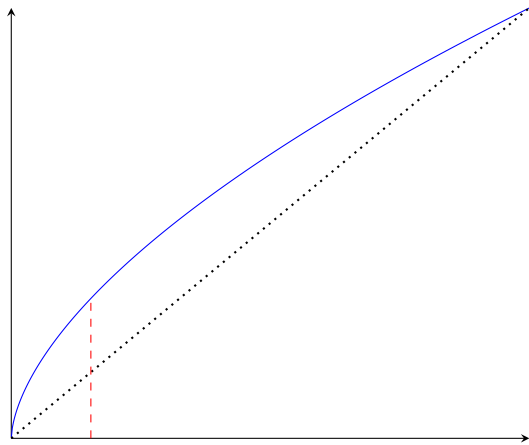
- q : 51.434766
- $\hat{q}^\beta$: 6.275281
- q^β : 10.635406
- Breakpoints: [0, 192.3352941176471]
- Error: 4.360124749395095
- Relative error: 0.409963182298349
- $cu, 1$
 - q : 7.527593
 - $\hat{q}^\beta$: 2.156788
 - q^β : 3.357327
 - Breakpoints: [0, 22.75735506822926]
 - Error: 1.200539481040013
 - Relative error: 0.357587860791341
- $cu, 3$
 - q : 7.499856
 - $\hat{q}^\beta$: 2.948352
 - q^β : 3.349899
 - Breakpoints: [0, 10.320104112088831, 128.44338870924773]
 - Error: 0.401547538225268
 - Relative error: 0.119868535642918
- $cu, 5$
 - q : 9.832104
 - $\hat{q}^\beta$: 3.356847
 - q^β : 3.940832
 - Breakpoints: [0, 14.682070969056184, 127.1036185995295]
 - Error: 0.583984718318929
 - Relative error: 0.148188195602391



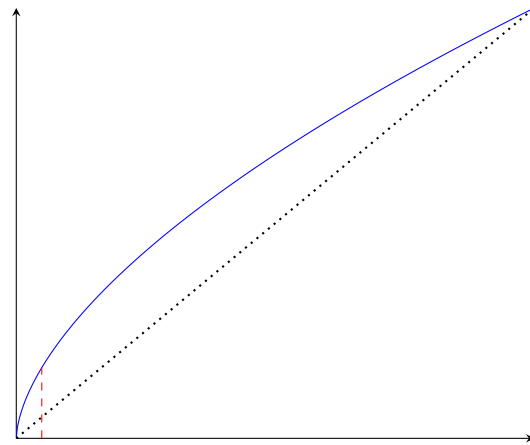
(a) $q^\beta - (2, 2, 2)$



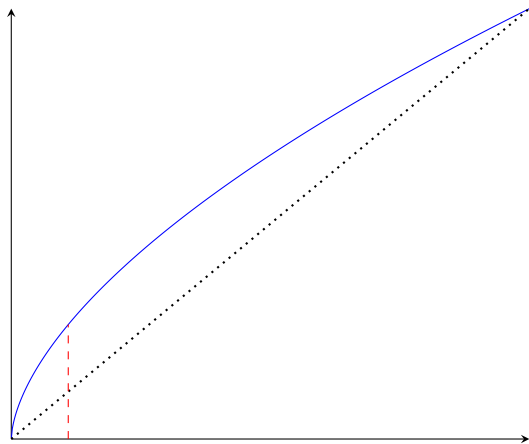
(b) $q^\beta - (2, 5, 2)$



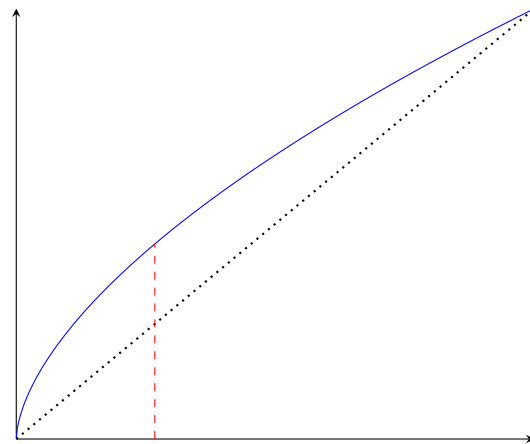
(c) $q^\beta - (3, 1, 2)$



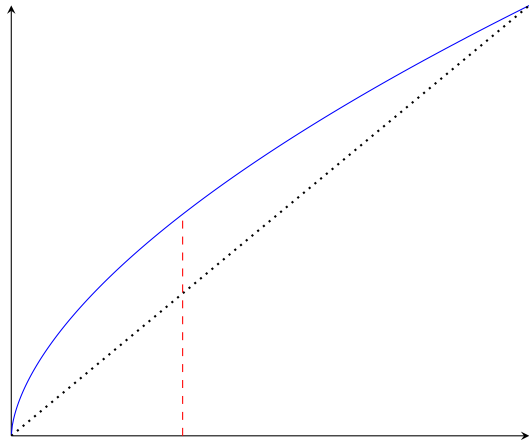
(d) $q^\beta - (3, 5, 2)$



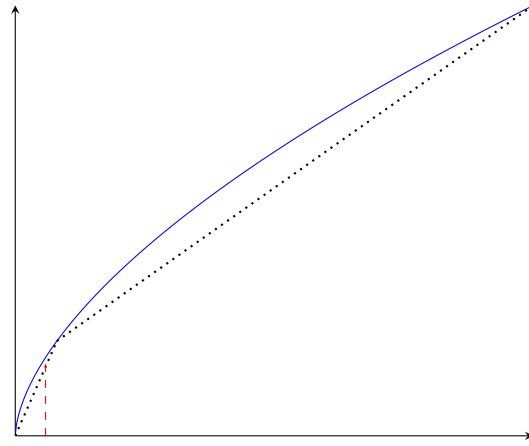
(e) $q^\beta - (4, 3, 2)$



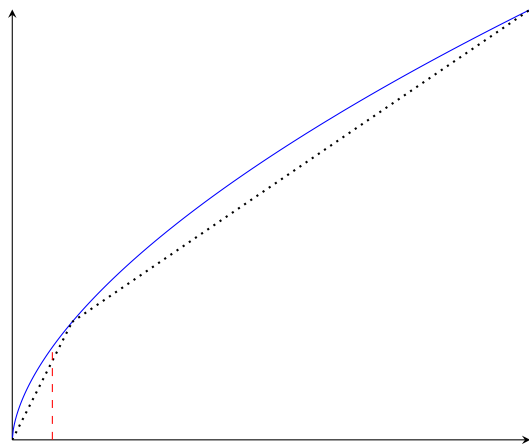
(f) $q^\beta - (5, 4, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 3$



(c) $q^\beta - cu, 5$

7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	0.0	$z_{4,5,1}$	0.0	$A_{2,4,1}^\beta$	0.0
$z_{1,1,2}$	-0.0	$z_{4,5,2}$	0.0	$A_{2,4,2}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{5,1,1}$	-0.0	$A_{2,5,1}^\beta$	0.0
$z_{1,2,2}$	-0.0	$z_{5,1,2}$	0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,2,1}$	0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,3,2}$	-0.0	$z_{5,2,2}$	-0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,3,1}$	0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,4,2}$	-0.0	$z_{5,3,2}$	-0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,4,1}$	0.0	$A_{3,4,1}^\beta$	0.0
$z_{1,5,2}$	-0.0	$z_{5,5,1}$	0.0	$A_{3,4,2}^\beta$	0.0
$z_{2,1,1}$	-0.0	$z_{5,5,2}$	-0.0	$A_{3,5,1}^\beta$	0.0
$z_{2,1,2}$	-0.0	$z_{cu,2}$	0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,2,1}$	0.0	$z_{cu,4}$	0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,3,1}$	0.0	$z_{hu,1}$	0.0	$A_{4,2,1}^\beta$	0.0
$z_{2,3,2}$	0.0	$z_{hu,2}$	0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,4,1}$	0.0	$z_{hu,3}$	0.0	$A_{4,3,1}^\beta$	0.0
$z_{2,4,2}$	0.0	$z_{hu,4}$	0.0	$A_{4,4,1}^\beta$	0.0
$z_{2,5,1}$	-0.0	$z_{hu,5}$	0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,1}$	-0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,1}^\beta$	0.0
$z_{3,2,1}$	-0.0	$A_{1,1,2}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,2,2}$	-0.0	$A_{1,2,1}^\beta$	0.0	$A_{5,1,1}^\beta$	0.0
$z_{3,3,1}$	0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,3,2}$	-0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,4,1}$	-0.0	$A_{1,3,2}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,4,2}$	-0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,5,1}$	-0.0	$A_{1,4,2}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{4,1,1}$	0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,4,1}^\beta$	0.0
$z_{4,1,2}$	0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,2,1}$	0.0	$A_{2,1,1}^\beta$	0.0	$A_{5,5,2}^\beta$	0.0
$z_{4,2,2}$	0.0	$A_{2,1,2}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,3,1}$	0.0	$A_{2,2,1}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,4,1}$	0.0	$A_{2,3,1}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,4,2}$	-0.0	$A_{2,3,2}^\beta$	0.0	$A_{hu,2}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,3}^\beta$	0.0	$A_{4,3,1}$	0.0	$dt_{1,5,1}$	66.7
$A_{hu,4}^\beta$	0.0	$A_{4,4,1}$	0.0	$dt_{1,5,2}$	66.7
$A_{hu,5}^\beta$	0.0	$A_{4,4,2}$	0.0	$dt_{1,5,3}$	66.7
$A_{1,1,1}$	0.0	$A_{4,5,1}$	0.0	$dt_{2,1,1}$	111.1
$A_{1,1,2}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,1,2}$	111.1
$A_{1,2,1}$	0.0	$A_{5,1,1}$	0.0	$dt_{2,1,3}$	100.0
$A_{1,2,2}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,2,1}$	111.1
$A_{1,3,1}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,3,1}$	111.1
$A_{1,3,2}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,3,2}$	10.0
$A_{1,4,1}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,3,3}$	100.0
$A_{1,4,2}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,5,1}$	0.0	$A_{5,4,1}$	0.0	$dt_{2,4,2}$	10.0
$A_{1,5,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{2,4,3}$	55.6
$A_{2,1,1}$	0.0	$A_{5,5,2}$	0.0	$dt_{2,5,1}$	111.1
$A_{2,1,2}$	0.0	$A_{cu,2}$	0.0	$dt_{3,1,1}$	161.1
$A_{2,2,1}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,3,1}$	0.0	$A_{hu,1}$	0.0	$dt_{3,2,2}$	111.1
$A_{2,3,2}$	0.0	$A_{hu,2}$	0.0	$dt_{3,2,3}$	111.1
$A_{2,4,1}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,4,2}$	0.0	$A_{hu,4}$	0.0	$dt_{3,3,2}$	161.1
$A_{2,5,1}$	0.0	$A_{hu,5}$	0.0	$dt_{3,3,3}$	188.9
$A_{3,1,1}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,1}$	144.5
$A_{3,2,1}$	0.0	$dt_{1,1,2}$	66.7	$dt_{3,4,2}$	144.5
$A_{3,2,2}$	0.0	$dt_{1,1,3}$	100.0	$dt_{3,4,3}$	10.0
$A_{3,3,1}$	0.0	$dt_{1,2,1}$	17.15006133296989	$dt_{3,5,1}$	133.4
$A_{3,3,2}$	0.0	$dt_{1,2,2}$	17.15006133296989	$dt_{4,1,1}$	122.2
$A_{3,4,1}$	0.0	$dt_{1,2,3}$	17.15006133296989	$dt_{4,1,2}$	10.0
$A_{3,4,2}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,3}$	88.9
$A_{3,5,1}$	0.0	$dt_{1,3,2}$	66.7	$dt_{4,2,1}$	122.2
$A_{4,1,1}$	0.0	$dt_{1,3,3}$	122.2	$dt_{4,2,2}$	10.0
$A_{4,1,2}$	0.0	$dt_{1,4,1}$	66.7	$dt_{4,2,3}$	33.3
$A_{4,2,1}$	0.0	$dt_{1,4,2}$	66.7	$dt_{4,3,1}$	122.2
$A_{4,2,2}$	0.0	$dt_{1,4,3}$	77.8	$dt_{4,4,1}$	122.2

Variable	Value	Variable	Value	Variable	Value
$dt_{4,4,2}$	10.0	$f_{1,5,1}^H$	8.8	$f_{5,4,1}^H$	17.7
$dt_{4,4,3}$	94.5	$f_{1,5,2}^H$	8.8	$f_{5,5,1}^H$	0.0
$dt_{4,5,1}$	122.2	$f_{2,1,1}^H$	0.0	$f_{5,5,2}^H$	0.0
$dt_{4,5,2}$	10.0	$f_{2,1,2}^H$	0.0	$f_{1,1,1}^C$	0.0
$dt_{4,5,3}$	55.6	$f_{2,2,1}^H$	0.0	$f_{1,1,2}^C$	0.0
$dt_{5,1,1}$	133.3	$f_{2,3,1}^H$	10.6	$f_{1,2,1}^C$	0.0
$dt_{5,1,2}$	17.152375493600957	$f_{2,3,2}^H$	0.0	$f_{1,2,2}^C$	0.0
$dt_{5,1,3}$	17.152375493600953	$f_{2,4,1}^H$	0.0	$f_{1,3,1}^C$	2.8
$dt_{5,2,1}$	74.66004781348401	$f_{2,4,2}^H$	0.0	$f_{1,3,2}^C$	0.0
$dt_{5,2,2}$	10.0	$f_{2,5,1}^H$	0.0	$f_{1,4,1}^C$	0.0
$dt_{5,2,3}$	21.35698491649854	$f_{3,1,1}^H$	14.8	$f_{1,4,2}^C$	0.0
$dt_{5,3,1}$	133.3	$f_{3,2,1}^H$	0.0	$f_{1,5,1}^C$	6.95
$dt_{5,3,2}$	53.28651348124793	$f_{3,2,2}^H$	3.7891992551211517	$f_{1,5,2}^C$	0.0
$dt_{5,3,3}$	161.1	$f_{3,3,1}^H$	0.0	$f_{2,1,1}^C$	0.0
$dt_{5,4,1}$	116.7	$f_{3,3,2}^H$	0.0	$f_{2,1,2}^C$	0.0
$dt_{5,5,1}$	105.6	$f_{3,4,1}^H$	0.0	$f_{2,2,1}^C$	0.0
$dt_{5,5,2}$	105.6	$f_{3,4,2}^H$	0.0	$f_{2,3,1}^C$	0.0
$dt_{5,5,3}$	105.6	$f_{3,5,1}^H$	0.0	$f_{2,3,2}^C$	0.0
$dt_{cu,2}$	97.8	$f_{4,1,1}^H$	12.6	$f_{2,4,1}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,1,2}^H$	0.0	$f_{2,4,2}^C$	0.0
$dt_{hu,1}$	80.0	$f_{4,2,1}^H$	0.0	$f_{2,5,1}^C$	0.0
$dt_{hu,2}$	18.3	$f_{4,2,2}^H$	0.0	$f_{3,1,1}^C$	0.0
$dt_{hu,3}$	18.9	$f_{4,3,1}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,4,1}^H$	0.0	$f_{3,2,2}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,4,2}^H$	0.0	$f_{3,3,1}^C$	2.8
$f_{1,1,1}^H$	0.0	$f_{4,5,1}^H$	0.0	$f_{3,3,2}^C$	0.0
$f_{1,1,2}^H$	0.0	$f_{4,5,2}^H$	0.0	$f_{3,4,1}^C$	17.3
$f_{1,2,1}^H$	0.0	$f_{5,1,1}^H$	0.0	$f_{3,4,2}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,1,2}^H$	0.0	$f_{3,5,1}^C$	0.0
$f_{1,3,1}^H$	0.0	$f_{5,2,1}^H$	0.0	$f_{4,1,1}^C$	7.6
$f_{1,3,2}^H$	0.0	$f_{5,2,2}^H$	0.0	$f_{4,1,2}^C$	0.0
$f_{1,4,1}^H$	0.0	$f_{5,3,1}^H$	0.0	$f_{4,2,1}^C$	6.1
$f_{1,4,2}^H$	0.0	$f_{5,3,2}^H$	5.435558889722431	$f_{4,2,2}^C$	0.0

Variable	Value	Variable	Value	Variable	Value
$f_{4,3,1}^C$	0.0	$\text{RecLMTD}_{3,2,1}$	0.1	$q_{1,1,2}$	0.0
$f_{4,4,1}^C$	0.0	$\text{RecLMTD}_{3,2,2}$	0.1	$q_{1,2,1}$	0.0
$f_{4,4,2}^C$	0.0	$\text{RecLMTD}_{3,3,1}$	0.1	$q_{1,2,2}$	0.0
$f_{4,5,1}^C$	0.0	$\text{RecLMTD}_{3,3,2}$	0.026431416364925124	$q_{1,3,1}$	0.0
$f_{4,5,2}^C$	0.0	$\text{RecLMTD}_{3,4,1}$	0.028945360806037865	$q_{1,3,2}$	0.0
$f_{5,1,1}^C$	0.0	$\text{RecLMTD}_{3,4,2}$	0.019856464048761648	$q_{1,4,1}$	0.0
$f_{5,1,2}^C$	0.0	$\text{RecLMTD}_{3,5,1}$	0.013119967177547875	$q_{1,4,2}$	0.0
$f_{5,2,1}^C$	0.0	$\text{RecLMTD}_{4,1,1}$	0.0203509210414817	$q_{1,5,1}$	0.0
$f_{5,2,2}^C$	0.0	$\text{RecLMTD}_{4,1,2}$	0.021502123357877247	$q_{1,5,2}$	0.0
$f_{5,3,1}^C$	2.8	$\text{RecLMTD}_{4,2,1}$	0.027922618193149052	$q_{2,1,1}$	0.0
$f_{5,3,2}^C$	0.0	$\text{RecLMTD}_{4,2,2}$	0.029297885186065315	$q_{2,1,2}$	0.0
$f_{5,4,1}^C$	0.0	$\text{RecLMTD}_{4,3,1}$	0.1	$q_{2,2,1}$	0.0
$f_{5,5,1}^C$	6.95	$\text{RecLMTD}_{4,4,1}$	0.02230651890829313	$q_{2,3,1}$	0.0
$f_{5,5,2}^C$	0.0	$\text{RecLMTD}_{4,4,2}$	0.024030191698075014	$q_{2,3,2}$	0.0
$\text{RecLMTD}_{1,1,1}$	0.1	$\text{RecLMTD}_{4,5,1}$	0.024784155348990783	$q_{2,4,1}$	0.0
$\text{RecLMTD}_{1,1,2}$	0.011664999999999987	$\text{RecLMTD}_{4,5,2}$	0.025546702599208902	$q_{2,4,2}$	0.0
$\text{RecLMTD}_{1,2,1}$	0.1	$\text{RecLMTD}_{5,1,1}$	0.030663111933576022	$q_{2,5,1}$	0.0
$\text{RecLMTD}_{1,2,2}$	0.1	$\text{RecLMTD}_{5,1,2}$	0.028476245063990456	$q_{3,1,1}$	0.0
$\text{RecLMTD}_{1,3,1}$	0.1	$\text{RecLMTD}_{5,2,1}$	0.1	$q_{3,2,1}$	0.0
$\text{RecLMTD}_{1,3,2}$	0.030706418976158202	$\text{RecLMTD}_{5,2,2}$	0.043215075417507326	$q_{3,2,2}$	0.0
$\text{RecLMTD}_{1,4,1}$	0.1	$\text{RecLMTD}_{5,3,1}$	0.1	$q_{3,3,1}$	0.0
$\text{RecLMTD}_{1,4,2}$	0.013770395384645764	$\text{RecLMTD}_{5,3,2}$	0.02882559254143146	$q_{3,3,2}$	0.0
$\text{RecLMTD}_{1,5,1}$	0.1	$\text{RecLMTD}_{5,4,1}$	0.1	$q_{3,4,1}$	0.0
$\text{RecLMTD}_{1,5,2}$	0.1	$\text{RecLMTD}_{5,5,1}$	0.1	$q_{3,4,2}$	0.0
$\text{RecLMTD}_{2,1,1}$	0.008877382668528087	$\text{RecLMTD}_{5,5,2}$	0.1	$q_{3,5,1}$	0.0
$\text{RecLMTD}_{2,1,2}$	0.02333947805307216	$\text{RecLMTD}_{cu,2}$	0.009423141115769088	$q_{4,1,1}$	0.0
$\text{RecLMTD}_{2,2,1}$	0.1	$\text{RecLMTD}_{cu,4}$	0.00853530405247822	$q_{4,1,2}$	0.0
$\text{RecLMTD}_{2,3,1}$	0.02467052252597492	$\text{RecLMTD}_{hu,1}$	0.012387051800747544	$q_{4,2,1}$	0.0
$\text{RecLMTD}_{2,3,2}$	0.02092652219967947	$\text{RecLMTD}_{hu,2}$	0.06086638680389446	$q_{4,2,2}$	0.0
$\text{RecLMTD}_{2,4,1}$	0.025165980106347283	$\text{RecLMTD}_{hu,3}$	0.06659303416456284	$q_{4,3,1}$	0.0
$\text{RecLMTD}_{2,4,2}$	0.026432601068855886	$\text{RecLMTD}_{hu,4}$	0.01543447453012974	$q_{4,4,1}$	0.0
$\text{RecLMTD}_{2,5,1}$	0.012806599920651493	$\text{RecLMTD}_{hu,5}$	0.024141568940966474	$q_{4,4,2}$	0.0
$\text{RecLMTD}_{3,1,1}$	0.02558564367826279	$q_{1,1,1}$	0.0	$q_{4,5,1}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{4,5,2}$	0.0	$t_{2,4,2}^H$	248.9	$t_{1,3,2}^C$	37.8
$q_{5,1,1}$	0.0	$t_{2,5,1}^H$	248.9	$t_{1,4,1}^C$	176.7
$q_{5,1,2}$	0.0	$t_{3,1,1}^H$	226.7	$t_{1,4,2}^C$	176.7
$q_{5,2,1}$	0.0	$t_{3,2,1}^H$	226.7	$t_{1,5,1}^C$	148.85
$q_{5,2,2}$	0.0	$t_{3,2,2}^H$	226.7	$t_{1,5,2}^C$	93.3
$q_{5,3,1}$	0.0	$t_{3,3,1}^H$	226.7	$t_{2,1,1}^C$	99.59210526315772
$q_{5,3,2}$	0.0	$t_{3,3,2}^H$	65.6	$t_{2,1,2}^C$	60.0
$q_{5,4,1}$	0.0	$t_{3,4,1}^H$	226.7	$t_{2,2,1}^C$	221.7
$q_{5,5,1}$	0.0	$t_{3,4,2}^H$	226.7	$t_{2,3,1}^C$	221.1
$q_{5,5,2}$	0.0	$t_{3,5,1}^H$	226.7	$t_{2,3,2}^C$	37.8
$q_{cu,2}$	0.0	$t_{4,1,1}^H$	271.1	$t_{2,4,1}^C$	82.2
$q_{cu,4}$	0.0	$t_{4,1,2}^H$	271.1	$t_{2,4,2}^C$	82.2
$q_{hu,1}$	0.0	$t_{4,2,1}^H$	271.1	$t_{2,5,1}^C$	204.4
$q_{hu,2}$	0.0	$t_{4,2,2}^H$	271.1	$t_{3,1,1}^C$	160.0
$q_{hu,3}$	0.0	$t_{4,3,1}^H$	271.1	$t_{3,2,1}^C$	221.7
$q_{hu,4}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,2,2}^C$	115.6
$q_{hu,5}$	0.0	$t_{4,4,2}^H$	271.1	$t_{3,3,1}^C$	111.5
$t_{1,1,1}^H$	160.0	$t_{4,5,1}^H$	271.1	$t_{3,3,2}^C$	37.8
$t_{1,1,2}^H$	93.3	$t_{4,5,2}^H$	271.1	$t_{3,4,1}^C$	176.7
$t_{1,2,1}^H$	160.0	$t_{5,1,1}^H$	198.9	$t_{3,4,2}^C$	82.2
$t_{1,2,2}^H$	160.0	$t_{5,1,2}^H$	126.72824858757069	$t_{3,5,1}^C$	204.4
$t_{1,3,1}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{4,1,1}^C$	160.0
$t_{1,3,2}^H$	160.0	$t_{5,2,2}^H$	65.6	$t_{4,1,2}^C$	60.0
$t_{1,4,1}^H$	160.0	$t_{5,3,1}^H$	198.9	$t_{4,2,1}^C$	221.7
$t_{1,4,2}^H$	160.0	$t_{5,3,2}^H$	106.53559322033877	$t_{4,2,2}^C$	115.6
$t_{1,5,1}^H$	160.0	$t_{5,4,1}^H$	198.9	$t_{4,3,1}^C$	221.1
$t_{1,5,2}^H$	160.0	$t_{5,5,1}^H$	198.9	$t_{4,4,1}^C$	82.2
$t_{2,1,1}^H$	198.8575471698114	$t_{5,5,2}^H$	65.6	$t_{4,4,2}^C$	82.2
$t_{2,1,2}^H$	248.9	$t_{1,1,1}^C$	160.0	$t_{4,5,1}^C$	204.4
$t_{2,2,1}^H$	248.9	$t_{1,1,2}^C$	160.0	$t_{4,5,2}^C$	93.3
$t_{2,3,1}^H$	248.9	$t_{1,2,1}^C$	221.7	$t_{5,1,1}^C$	60.0
$t_{2,3,2}^H$	137.8	$t_{1,2,2}^C$	115.6	$t_{5,1,2}^C$	60.0
$t_{2,4,1}^H$	248.9	$t_{1,3,1}^C$	111.5	$t_{5,2,1}^C$	221.7

Variable	Value	Variable	Value	Variable	Value
$t_{5,2,2}^C$	115.6	$b_{4,1,2}^{H,in}$	0.0	$b_{2,4,2}^{H,out}$	0.0
$t_{5,3,1}^C$	111.5	$b_{4,2,1}^{H,in}$	0.0	$b_{2,5,1}^{H,out}$	0.0
$t_{5,3,2}^C$	37.8	$b_{4,2,2}^{H,in}$	0.0	$b_{3,1,1}^{H,out}$	3355.16
$t_{5,4,1}^C$	176.7	$b_{4,3,1}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	0.0
$t_{5,5,1}^C$	148.85	$b_{4,4,1}^{H,in}$	0.0	$b_{3,2,2}^{H,out}$	859.011471135965
$t_{5,5,2}^C$	93.3	$b_{4,4,2}^{H,in}$	0.0	$b_{3,3,1}^{H,out}$	0.0
$b_{1,1,1}^{H,in}$	0.0	$b_{4,5,1}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	0.0
$b_{1,1,2}^{H,in}$	0.0	$b_{4,5,2}^{H,in}$	0.0	$b_{3,4,1}^{H,out}$	0.0
$b_{1,2,1}^{H,in}$	0.0	$b_{5,1,1}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	0.0
$b_{1,2,2}^{H,in}$	0.0	$b_{5,1,2}^{H,in}$	0.0	$b_{3,5,1}^{H,out}$	0.0
$b_{1,3,1}^{H,in}$	0.0	$b_{5,2,1}^{H,in}$	0.0	$b_{4,1,1}^{H,out}$	3415.86
$b_{1,3,2}^{H,in}$	0.0	$b_{5,2,2}^{H,in}$	0.0	$b_{4,1,2}^{H,out}$	0.0
$b_{1,4,1}^{H,in}$	0.0	$b_{5,3,1}^{H,in}$	0.0	$b_{4,2,1}^{H,out}$	0.0
$b_{1,4,2}^{H,in}$	0.0	$b_{5,3,2}^{H,in}$	1081.1326631657917	$b_{4,2,2}^{H,out}$	0.0
$b_{1,5,1}^{H,in}$	1408.0	$b_{5,4,1}^{H,in}$	3520.5299999999997	$b_{4,3,1}^{H,out}$	0.0
$b_{1,5,2}^{H,in}$	1408.0	$b_{5,5,1}^{H,in}$	0.0	$b_{4,4,1}^{H,out}$	0.0
$b_{2,1,1}^{H,in}$	0.0	$b_{5,5,2}^{H,in}$	0.0	$b_{4,4,2}^{H,out}$	0.0
$b_{2,1,2}^{H,in}$	0.0	$b_{1,1,1}^{H,out}$	0.0	$b_{4,5,1}^{H,out}$	0.0
$b_{2,2,1}^{H,in}$	0.0	$b_{1,1,2}^{H,out}$	0.0	$b_{4,5,2}^{H,out}$	0.0
$b_{2,3,1}^{H,in}$	2638.34	$b_{1,2,1}^{H,out}$	0.0	$b_{5,1,1}^{H,out}$	0.0
$b_{2,3,2}^{H,in}$	0.0	$b_{1,2,2}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	0.0
$b_{2,4,1}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	0.0
$b_{2,4,2}^{H,in}$	0.0	$b_{1,3,2}^{H,out}$	0.0	$b_{5,2,2}^{H,out}$	0.0
$b_{2,5,1}^{H,in}$	0.0	$b_{1,4,1}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	0.0
$b_{3,1,1}^{H,in}$	3355.16	$b_{1,4,2}^{H,out}$	0.0	$b_{5,3,2}^{H,out}$	1081.1326631657919
$b_{3,2,1}^{H,in}$	0.0	$b_{1,5,1}^{H,out}$	1408.0	$b_{5,4,1}^{H,out}$	3520.5299999999997
$b_{3,2,2}^{H,in}$	859.011471135965	$b_{1,5,2}^{H,out}$	1408.0	$b_{5,5,1}^{H,out}$	0.0
$b_{3,3,1}^{H,in}$	0.0	$b_{2,1,1}^{H,out}$	0.0	$b_{5,5,2}^{H,out}$	0.0
$b_{3,3,2}^{H,in}$	0.0	$b_{2,1,2}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,4,1}^{H,in}$	0.0	$b_{2,2,1}^{H,out}$	0.0	$b_{1,1,2}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	0.0	$b_{2,3,1}^{H,out}$	2638.34	$b_{1,2,1}^{C,in}$	0.0
$b_{3,5,1}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	0.0	$b_{1,2,2}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	3415.86	$b_{2,4,1}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	619.08

Variable	Value	Variable	Value	Variable	Value
$b_{1,3,2}^{C,in}$	0.0	$b_{5,2,2}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	0.0
$b_{1,4,1}^{C,in}$	0.0	$b_{5,3,1}^{C,in}$	619.08	$b_{4,2,1}^{C,out}$	1352.37
$b_{1,4,2}^{C,in}$	0.0	$b_{5,3,2}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	0.0
$b_{1,5,1}^{C,in}$	1420.5800000000002	$b_{5,4,1}^{C,in}$	0.0	$b_{4,3,1}^{C,out}$	0.0
$b_{1,5,2}^{C,in}$	0.0	$b_{5,5,1}^{C,in}$	1420.5800000000002	$b_{4,4,1}^{C,out}$	0.0
$b_{2,1,1}^{C,in}$	0.0	$b_{5,5,2}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	0.0
$b_{2,1,2}^{C,in}$	0.0	$b_{1,1,1}^{C,out}$	0.0	$b_{4,5,1}^{C,out}$	0.0
$b_{2,2,1}^{C,in}$	0.0	$b_{1,1,2}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	0.0
$b_{2,3,1}^{C,in}$	0.0	$b_{1,2,1}^{C,out}$	0.0	$b_{5,1,1}^{C,out}$	0.0
$b_{2,3,2}^{C,in}$	0.0	$b_{1,2,2}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{2,4,1}^{C,in}$	0.0	$b_{1,3,1}^{C,out}$	619.08	$b_{5,2,1}^{C,out}$	0.0
$b_{2,4,2}^{C,in}$	0.0	$b_{1,3,2}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{2,5,1}^{C,in}$	0.0	$b_{1,4,1}^{C,out}$	0.0	$b_{5,3,1}^{C,out}$	619.08
$b_{3,1,1}^{C,in}$	0.0	$b_{1,4,2}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{3,2,1}^{C,in}$	0.0	$b_{1,5,1}^{C,out}$	1420.5800000000002	$b_{5,4,1}^{C,out}$	0.0
$b_{3,2,2}^{C,in}$	0.0	$b_{1,5,2}^{C,out}$	0.0	$b_{5,5,1}^{C,out}$	1420.5800000000002
$b_{3,3,1}^{C,in}$	619.08	$b_{2,1,1}^{C,out}$	0.0	$b_{5,5,2}^{C,out}$	0.0
$b_{3,3,2}^{C,in}$	0.0	$b_{2,1,2}^{C,out}$	0.0		
$b_{3,4,1}^{C,in}$	3056.91	$b_{2,2,1}^{C,out}$	0.0		
$b_{3,4,2}^{C,in}$	0.0	$b_{2,3,1}^{C,out}$	0.0		
$b_{3,5,1}^{C,in}$	0.0	$b_{2,3,2}^{C,out}$	0.0		
$b_{4,1,1}^{C,in}$	1216.0	$b_{2,4,1}^{C,out}$	0.0		
$b_{4,1,2}^{C,in}$	0.0	$b_{2,4,2}^{C,out}$	0.0		
$b_{4,2,1}^{C,in}$	1352.37	$b_{2,5,1}^{C,out}$	0.0		
$b_{4,2,2}^{C,in}$	0.0	$b_{3,1,1}^{C,out}$	0.0		
$b_{4,3,1}^{C,in}$	0.0	$b_{3,2,1}^{C,out}$	0.0		
$b_{4,4,1}^{C,in}$	0.0	$b_{3,2,2}^{C,out}$	0.0		
$b_{4,4,2}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	619.08		
$b_{4,5,1}^{C,in}$	0.0	$b_{3,3,2}^{C,out}$	0.0		
$b_{4,5,2}^{C,in}$	0.0	$b_{3,4,1}^{C,out}$	3056.91		
$b_{5,1,1}^{C,in}$	0.0	$b_{3,4,2}^{C,out}$	0.0		
$b_{5,1,2}^{C,in}$	0.0	$b_{3,5,1}^{C,out}$	0.0		
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	1216.0		